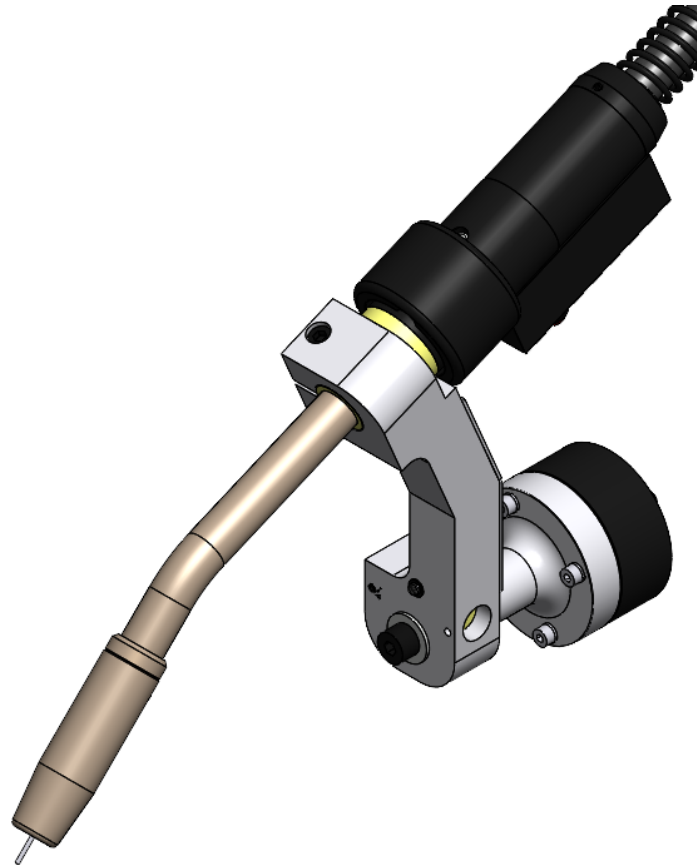




**Robotic Collaborative MIG Torches
For Models
300 Amp, 500 Amp Air-Cooled & 550 Amp
Water-Cooled**



This Product is
MADE IN THE U.S.A.

INSTALLATION, OPERATIONS, AND REPLACEMENT PARTS

SERVICE QUALITY SOLUTIONS

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INTRODUCTION

Thank you for purchasing an **AMERICAN WELDDQUIP** product. The **AMERICAN WELDDQUIP** product you have purchased has been carefully manufactured, assembled, and fully tested. This manual contains information on the installation, operation, maintenance, and replacement part breakdown. Please read, understand, and follow all safety instructions, warnings, and procedures. Keep this manual handy for referencing installation, operation, maintenance, and part ordering information. While every precaution has been taken as to the accuracy in this manual, **AMERICAN WELDDQUIP** assumes no responsibility for errors or omissions. **AMERICAN WELDDQUIP** assumes no liability for damages resulting from the use of the information contained in this manual. **AMERICAN WELDDQUIP** shall have no liability to the buyer for consequential damages or expenses by any defect whatsoever.

WARRANTY

AMERICAN WELDDQUIP MIG guns and parts are warranted to be free of defects in material and/or workmanship for the period listed below. For any product found to be defective under normal use, **AMERICAN WELDDQUIP** at our option, will repair, replace, or issue a credit for the value of the defective product. All warranty claims must be submitted by the original purchaser. Use of non-genuine **AMERICAN WELDDQUIP** parts and/or consumables may damage and/or severely limit the performance of the equipment which may limit or void any warranties. **AMERICAN WELDDQUIP** will not assume responsibility for incidental damages or expenses related to any defect. This warranty does not cover damage caused by misuse or abuse, accident, alteration of product, improper installation, misapplication, lack of reasonable care and maintenance, unauthorized repairs or modifications, loss of use while at a repair facility or other conditions that are beyond the control of **AMERICAN WELDDQUIP**.

A Return Merchandise Authorization Number (RMA#) must be attained from the factory for any product being returned for Warranty Repair or Replacement. All returned products must be shipped freight prepaid by the sender. No-charge replacements, repaired products, or credit will be issued once the returned product has been evaluated and warranty condition has been verified. If an immediate replacement is required before proper warranty evaluation, a purchase order number is required, and the goods will be invoiced. A credit will be issued once it is determined that a warranty condition exists.

STANDARD WARRANTY

All Semi-Automatic, Automatic, Robotic MIG Torches and Components.....= 120 Days
MIG Torch Trigger Switches (Contacts only) - Excludes Smoke Extraction.....= LIFETIME
Robotic Nozzle Cleaning Stations, Wire Cutter.....= 6 Months
Robotic Peripherals, ArcSafe™, Gun Mounts.....= 90 Days
TIG POINT Tungsten Electrode Grinders.....= 90 Days

LIMITED EXTENDED WARRANTY PROTECTION

This limited extended warranty protection expands coverage to loyal customers who use all **GENUINE AMERICAN WELDDQUIP** consumables. Customers filing a claim under the extended warranty will need to prove, by providing past invoices, that they have been purchasing and using **Genuine AMERICAN WELDDQUIP** consumables.

All Semi-Automatic, Automatic, Robotic MIG Torches and Components.....= 1 Year
MIG Torch Trigger Switches (Contacts only) -Excludes Smoke Extraction & Pistol... = LIFETIME
MIG Torch Handles..... = LIFETIME
Robotic Nozzle Cleaning Stations, Wire Cutter.....= 1 Year w/Exclusive Quip-Mist Use
Robotic Peripherals, ArcSafe™, Gun Mounts.....= 90 Days

ROHS COMPLIANT

RoHS (Restriction of Hazardous Substances) is an environmental law which addresses the European Union directive 2002/95/EC known as the RoHS Directive. The RoHS directive restricts the use of hazardous substances listed below in electrical and electronic equipment. While it is not a requirement to meet the directive in the United States, currently, **AMERICAN WELDQUIP INC.** feels this is an important part of our “Go Green initiative. We have taken all reasonable steps to try to ensure the supporting evidence regarding the absence of the restricted substances to support RoHS compliance.

For reference, the maximum concentration values of the restricted substances by weight in homogenous materials are:

Lead/Lead Components.....	-0.1%
Mercury.....	-0.1%
Hexavalent Chromium.....	-0.1%
Polybrominated Biphenyls (PBBs).....	-0.1%
Polybrominated Diphenyl Ethers (PBDEs).....	-0.1%
Cadmium.....	-0.01%

For RoHS Certification of Compliance Letter on a particular product please visit our website – www.weldquip.com or email us at technical@weldquip.com or call 330-239-0317.



SAFETY PRECAUTIONS – READ BEFORE USING

Before installing, operating, or performing maintenance please read the safety precautions below. Failure to observe safety precautions can result in injury or death.

Read and follow the Owner's Manual carefully before installing, operating, or servicing equipment. Read and understand all safety information.

CALIFORNIA PROPOSITION 65 WARNINGS

This product, when used for welding and cutting, can produce fumes or gases which contain chemicals known to cause birth defects and cancer. (California Health & Safety Code Section 25249.5 et seq.) **WASH HANDS AFTER HANDLING.**

EMF – ELECTRICAL AND MAGNETIC FIELDS MAY BE DANGEROUS

Electrical current flowing through any conductor causes localized Electric and Magnetic Fields (EMF). Welding current creates an EMF field around welding cables and welding machines.

WARNING - EMF fields may interfere with some pacemakers and other medical implants. Implanted medical device wearers should consult their doctor before operating or going near any arc welding applications. In addition, exposure to EMF fields in welding may have other unknown health effects.

- Welders should use the below procedures to minimize the exposure to EMF fields from the welding circuit.
- Route the cables close together. Secure by twisting, taping, or using a cable cover to keep together.
- Never coil, wrap, or drape welding cables around your body.
- Do not place your body between welding cables. Arrange it so that cables are on one side and away from the operator.
- Connect the work clamp (ground) to the workpiece as close as possible to the area to be welded.
- Do not sit, lean, or stand next to the welding power source.
- Do not breathe the fumes and gases as they can cause asphyxiation.

FUMES AND GASES CAN BE DANGEROUS



WARNING - WELDING AND CUTTING PRODUCE FUMES AND GASES THAT ARE HAZARDOUS TO YOUR HEALTH

Fumes and gases generated from welding can cause severe injury to respiratory system and even death. In poorly vented areas it is required to properly ventilate the area and/or use local forced ventilation or other fume control equipment at the arc to remove welding and cutting fumes and gases.

The recommended way to determine adequate ventilation is to sample the composition and quantity of fumes and gases to which personnel are exposed. The worker exposure level should be checked initially and periodically thereafter to maintain applicable OSHA PEL and ACGIH TVL limits.

- In a poorly ventilated area, it is necessary to wear an approved air-supplied respirator.
- Always read and understand the Safety Data Sheets (SDSs) and the manufacturer's instructions for adhesives, coatings, cleaners, consumables, coolants, degreasers, fluxes, and metals.

- Always have a trained watchperson nearby. Welding and cutting fumes and gases can displace air and lower the oxygen level causing injury or death. Be sure the breathing air is safe.
- Do not weld or cut in locations near degreasing, cleaning, or spraying operations. The heat and rays of the arc can react with vapors to form highly toxic and irritating gases.
- Do not weld or cut on coated metals, such as galvanized, lead, or cadmium plated steel, unless the coating is removed from the weld area, the area is well ventilated, and while wearing an air-supplied respirator. The coatings and any metals containing these elements can give off toxic fumes if welded.

ELECTRIC SHOCK CAN KILL



WARNING - ELECTRICAL SHOCK CAN KILL. DO NOT TOUCH LIVE ELECTRICAL PARTS AND/OR USE IN DAMP LOCATIONS.

The electrode and work (ground) circuit are electrically “HOT” whenever the welding equipment is on. Do not touch these electrically live parts with your bare skin or wet/damp clothing. Wear dry, hole-free gloves. Incorrectly installed or improperly grounded equipment is a hazard.

Insulate yourself from work and the ground using dry insulating mats or covers big enough to prevent any physical contact with the work or ground.

Additional safety precautions are required when any of the following electrically hazardous conditions are present: in damp locations or while wearing wet clothing; on metal structures such as floors, gratings, or scaffolds; when in cramped positions such as sitting, kneeling, or lying; or when there is a high risk of unavoidable or accidental contact with the workpiece or ground. For these conditions, use the following equipment in order presented: 1) a semi-automatic DC constant voltage, 2) a DC manual (stick) welder or 3) an AC welder with reduced open-circuit voltage. In most situations, use of a DC constant voltage wire welder is recommended and DO NOT work alone!

- Disconnect input power or stop engine before installing or servicing this equipment. Lockout/tag out input power according to OSHA 29 CFR 1910.147 (see Safety Standards).
- Properly install ground and operate this equipment according to its Owner’s Manual and national, state/provincial, and local codes.
- Always verify the supply ground. Make sure that input power cord ground wire is properly connected to ground terminal in disconnect box or that cord plug is connected to a properly grounded receptacle outlet.
- Keep cords dry, free of oil and grease and protected from hot metal and sparks.
- Frequently inspect input power cord for damage or bare wiring. Replace the cord immediately if damaged. Bare wiring can kill.
- Turn off all equipment when not in use.
- Do not use worn, damaged, undersized, or poorly spliced cables. It is illegal to use electrical tape to repair torch power cable or ground cable that has damaged outer insulation. The cable must be replaced.
- Do not drape cables over your body.
- Do not touch the electrode if you are in contact with the work, ground, or another electrode from a different machine.
- Do not touch electrode holders connected to two welding machines at the same time since double open circuit voltage will be present.

- Use only well-maintained equipment. Repair or replace damaged parts at once. Maintain unit according to manual.
- Wear a safety harness if working above floor level.
Keep all panels and covers securely in place.
- Clamp work cable with good metal-to-metal contact to workpiece or worktable as near the weld as practical.
- Do not connect more than one electrode or work cable to any single weld output terminal. Disconnect cable when not in use.

ARC RAYS HAZARDS



WARNING – A WELDING ARC EMITS ULTRAVIOLET (UV) AND OTHER RADIATION AND CAN CAUSE SERIOUS INJURY TO UNPROTECTED SKIN AND EYES.

Use a shield with the proper filter and cover plates to protect your eyes from sparks and the rays of the arc when welding or observing open arc welding. The headshield and filter lens should conform to ANSI Z87.1

standards.

Use suitable clothing made from durable flame-resistant material to protect your skin and that of your helpers from the arc rays.

Protect other nearby personnel with suitable, non-flammable screening and/or warn them not to watch the arc nor expose themselves to the arc rays or to hot spatter or metal.

WELDING AND CUTTING CAN CAUSE FIRE OR EXPLOSION



WARNING – Welding and cutting produces sparks that fly off from the arc and can cause fires and/or explosions.

- Welding or cutting on closed containers, such as tanks, drums or pipes can cause them to blow up. Sparks can fly off from the welding or cutting arc. The flying sparks, hot work pieces and hot equipment can cause fires and burns.
- Accidental contact of electrode to metal objects can cause sparks, explosions, overheating or fire. Check and be sure the area is safe before doing any welding or cutting.
- Do not weld or cut where flying sparks can strike flammable material.
- Remove all flammables and fire hazards from the welding area. If this is not possible, tightly cover them with approved covers to prevent the welding sparks from starting a fire.
- When not welding, make certain no part of the electrode circuit is touching the work or ground. Accidental contact can cause overheating and create a fire hazard.
- Be alert that welding sparks and hot materials from welding and cutting can easily go through small cracks and openings and cause a fire in the adjacent areas.

Follow requirements in OSHA 1910.252 (a) (2) (iv) and NFPA 51B for hot work and have a fire watcher and extinguisher nearby.

- Do not heat, cut, or weld tanks, drums or containers that have held combustibles until the proper steps have been taken to ensure that such procedures will not cause flammable or toxic vapors from substances inside. They can cause an explosion even though they have been “cleaned.” For information, purchase “Recommended Safe Practices for the Preparation for Welding and Cutting of Containers and Piping That Have Held Hazardous Substances,” AWS F4.1 from the American Welding Society.

- Do not weld or cut where the atmosphere may contain flammable dust, gas, or liquid vapors (such as gasoline).
- Sparks and spatter are thrown from the welding arc. Wear oil free protective garments such as leather gloves, heavy shirt, cuff-less trousers, high shoes, and a cap over your hair. Wear ear plugs when welding out of position or in confined places. Always wear safety glasses with side shields when in a welding area.
- Connect work cable to the work as close to the welding or cutting area as practical to prevent welding or cutting current from traveling long, possibly unknown paths and causing electric shock, sparks, and fire hazards.
- Do not use a welder to thaw frozen pipes.
- Remove any combustibles, such as a butane lighter or matches, from your person before doing any welding or cutting.
- Inspect the area to ensure it is free of sparks, glowing embers, and flames after work is complete.

CYLINDERS CAN EXPLODE IF DAMAGED



WARNING – Compressed gas cylinders contain gas under high pressure and/or flammable gas. If damaged, the cylinder can explode.

- Use only compressed gas cylinders containing the correct shielding gas for the process used and properly operating regulators designed for the gas and pressure used. All hoses, fittings, etc. should be suitable for the application and maintained in good condition.
- Always keep cylinders in an upright position securely chained to an undercarriage or fixed support.
- Cylinders should be located away from areas where they may be struck or subjected to physical damage and a safe distance from arc welding or cutting operations and any other source of heat, sparks, or flame.
- Never allow the electrode, electrode holder or any other electrically “hot” parts to touch a cylinder.
- Keep your head and face away from the cylinder valve outlet when opening the cylinder valve.
- Valve protection caps should always be in place and hand tight except when the cylinder is in use or connected for use.
- Read and follow the instructions on compressed gas cylinders, associated equipment, and CGA publication P-1, “Precautions for Safe Handling of Compressed Gases in Cylinders,” available from the Compressed Gas Association, 14501 George Carter Way Chantilly, VA 20151.

PRINCIPAL SAFETY STANDARDS

- *Safety in Welding, Cutting and Allied Processes, ANSI Standard Z49.1* – available for download from the American Welding Society website – www.aws.org.
- *CSA Standard W117.2* – available from Canadian Standards Association, Standards Sales, 5060 Spectrum Way, Suite 100, Ontario, Canada L4W 5NS or website – www.csa-international.org.
- *Nation Electric Code, NFPA Standard 70* – available from National Fire Protection Association, Quincy, MA 02269, or website – www.nfpa.org.
- *Safe Practices for Occupational and Educational Eye and Face Protection, ANSI Standard Z87.1* – available from the American National Standards Institute, 25 West 43rd Street, New York, NY 10036. Website – www.ansi.org.
- *OSHA, Occupational Safety and Health Standard for General Industry, Title 29, Code of Federal Regulations, Part 1910, Subpart Q and Part 1926, Subpart J* – available from U.S. Government Printing Office, Superintendent of Documents, P.O. Box 371954, Pittsburg, PA 15250. Website – www.osha.gov.

FEATURES

AMERICAN WELDQUIP collaborative robot torches have been designed with advanced features and benefits to include:

Adaptability – Designed to adapt to current **AMERICAN WELDQUIP** Air and Water-Cooled Robotic torch systems.

Input Controls – Controls at the gooseneck for manual input while manipulating the torch in the work area to plot weld points.

Internal Control Leads – All wiring is internal to each component, so it is not exposed in the work area.

Lighting – Bright White LED to light up the work area when controls are active for easier viewing.

Durability – Heavy-walled gooseneck design helps with heat sink dissipation.

Robust Design – Provides unparalleled resistance to bending in “crash situations.”

Locating Flat – Unique locating flat machined into block to guarantee TCP repetition when changing out goosenecks.

Consumables – Common consumables across **AMERICAN WELDQUIP** torches alleviate inventory requirements.

Strain-Relief – Our spring strain-relief prevents the power cable from being over bent prolonging its lifetime while preventing feed issues.

Customs – Special designs and custom configurations and lengths are readily available.

Location – Manufactured in Sharon Center, Ohio allowing for quick turnaround.

TORCH SPECIFICATIONS

MODEL	AMPERAGE	100% Duty Cycle		60% Duty Cycle	
		CO2	Mixed Gas (AR/CO2)	CO2	Mixed Gas (AR/CO2)
300R	300A	300A	225A	350A	300A
500R	500A	500A	390A	550A	500A
550R	550A	550A	450A	625A	550A

Duty Cycle is based on a complete cycle time of 10 minutes. (60% Duty Cycle = 6 minutes weld time, 4 minutes off time).

INSTALLATION

Depending on how it is ordered, your **AMERICAN WELDEQUIP** torch has been supplied with either a EURO type feeder connection or a DIRECT wire feeder connection. The **AMERICAN WELDEQUIP** torch can be installed to the wire feed unit in two ways.

- Euro Connection (Feeder Adaptor Kit may be required)
- Direct Connection

EURO CONNECTION (FEEDER ADAPTOR KIT)

In some cases, it may be desirable to use a Euro feeder adaptor kit when using different manufactures wire feed units. This will commonize torches to a Euro style configuration.

- 1) Thread the feeder adaptor plug into the adaptor block and tighten.
- 2) Insert the adaptor guide tube into the adaptor plug.
- 3) Fully insert the feeder adaptor assembly into the wire feeder. Tighten screw or other method on wire feeder to secure the adaptor assembly.
- 4) Connect the Euro torch to the feeder adaptor assembly.
- 5) Feed welding wire into the torch and tighten drive rolls.

WARNING: WHEN FEEDING WELD WIRE THROUGH THE TORCH KEEP THE FRONT END OF THE TORCH POINTED AWAY FROM ANY PERSON OR OBJECT. DO NOT POINT AT THE FACE, HANDS ETC. FAILURE TO DO SO WILL RESULT IN BODILY INJURY AND POSSIBLY DEATH.

DIRECT CONNECT

The direct connect torch system is designed for installation to the wire feeder without the need for any adaptor system. The torch is supplied (depending on the torch configuration ordered) with a feeder connection plug at the rear of the torch, gas connection, and feeder control cable.

- 1) Fully insert the torch Direct Connection into wire feeder. Tighten screw or other method on wire feeder to secure torch.
- 2) Connect the gas hose if required:
 - a) Lincoln Universal – connect PtoC to feeder pin then hose from PtoC to feeder.
 - b) OTC/Daihen, Lincoln, and Panasonic – connect to feeder with attached hose.
- 3) Connect the power cable if required (Lincoln, NA5, and OTC/Daihen feeder pins).
- 4) Feed welding wire into the torch and tighten drive rolls.

WARNING: WHEN FEEDING WELD WIRE THROUGH THE TORCH, KEEP THE FRONT END OF THE TORCH POINTED AWAY FROM ANY PERSON OR OBJECT. DO NOT POINT AT THE FACE, HANDS ETC. FAILURE TO DO SO WILL RESULT IN BODILY INJURY AND POSSIBLY DEATH.

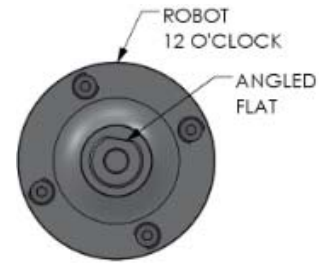
INSULATING DISC INSTALLATION

To ensure proper installation position of torch or other hardware, orientate torch in position with insulating disk before mounting. Otherwise, home position in relation to robot may not be correct.

- 1) Place the robot wrist face in the Home or Maintenance position.
- 2) Remove the Insulating Disc and mounting screws from the packaging.
- 3) Align the Insulating Disc with the holes on the robot wrist face.
- 4) Secure with the provided screws and torque to 45 in-lbs. (5 nm).

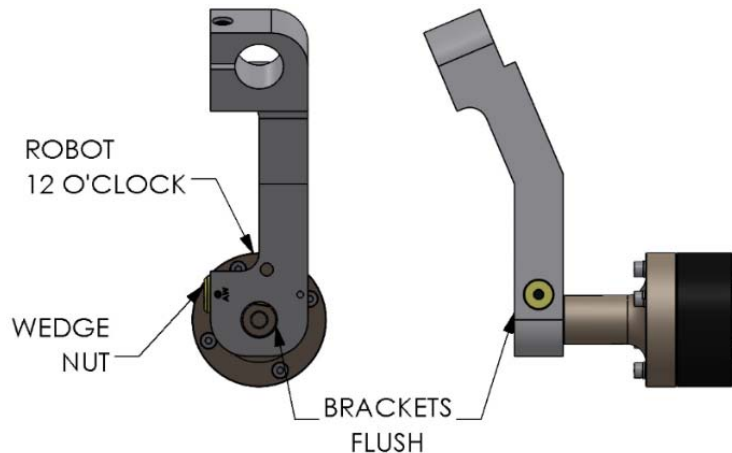
ARCSAFE™ SOLID MOUNT INSTALLATION

- 1) Remove ArcSafe™ Solid Mount from the packaging.
- 2) Align the ArcSafe™ Solid Mount to the Insulating Disc with the flat on the end facing up. The flat will be sloped to the right when looking from the end to the robot wrist face.
- 3) Secure with the provided screws and torque to 45 in-lbs. (5 nm).
- 4) Remove the Cap Screw and Washer from the end of the ArcSafe™ Solid Mount and set aside for reuse.



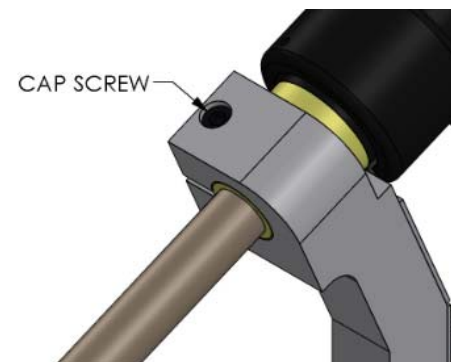
ARCSAFE™ COBOT BRACKET INSTALLATION

- 1) Remove ArcSafe™ Cobot Bracket from the packaging.
- 2) Loosen the screw holding the wedge nut in the bottom of the ArcSafe™ Cobot Bracket.
- 3) Orientate the wedge nut on the ArcSafe™ Cobot Bracket so it will slide onto the ArcSafe™ Solid Mount. Push back so the ArcSafe™ Cobot Bracket is flush to the front of the ArcSafe™ Solid Mount. The ArcSafe™ Cobot Bracket should be straight up (12 o'clock position) and bent forward (away from the robot wrist face).
- 4) Tighten the wedge nut to secure the ArcSafe™ Cobot Bracket to the ArcSafe™ Solid Mount.
- 5) Attach the Cap Screw and Washer to the end of the ArcSafe™ Solid Mount and tighten.

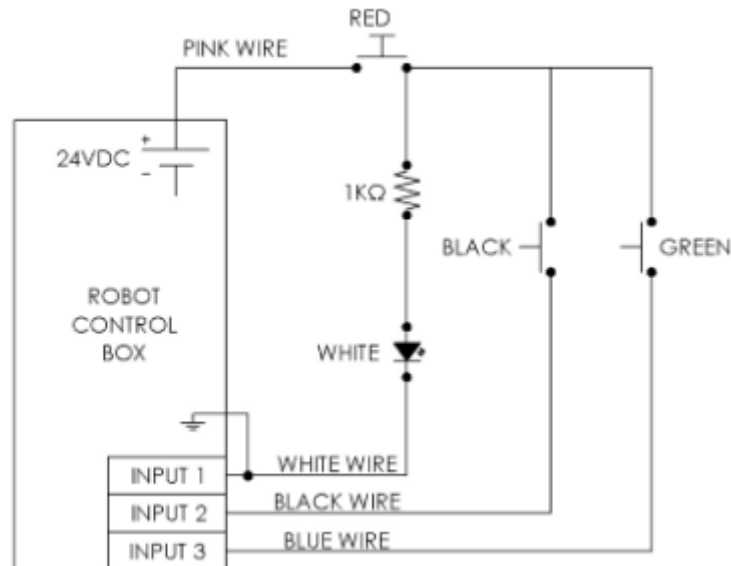


TORCH MOUNT INSTALLATION

- 1) Loosen the Cap Screw in the top of the ArcSafe™ Cobot Bracket.
- 2) Insert the Gooseneck of the Torch Assembly through the top of the mount aligning the flat on the Gooseneck with the flat on the ArcSafe™ Cobot Bracket.
- 3) Tighten the Cap Screw in the top of the ArcSafe™ Cobot Bracket to secure the Gooseneck.



ELECTRICAL SCHEMATIC



TORCH CONTROLS

- Red ON/OFF Switch (White wire) – Adds 24VDC to input Buttons and LED. Input buttons will not function unless turned ON.
- LED – Bright White LED lights up the work area at the tip of the torch. Turn on the Red ON/OFF Switch to activate the LED.
- Black Momentary Button (Black wire) – Inputs 24VDC to Robot Control Box. Can be used for multiple functions. Single press, Double press, Long press, etc.
- Green Momentary Button (Blue wire) – Inputs 24VDC to Robot Control Box. Can be used for multiple functions. Single press, Double press, Long press, etc.

ROBOT CONTROLS

- Pink Wire - Power to torch – up to 24VDC
- White Wire - Input 1 and/or ground – up to 24VDC signal from torch. Signal from Red ON/OFF Switch. Can also be parallel to ground or just ground if Input 1 is not needed.
- Black Wire - Input 2 – up to 24VDC signal from Black Button.
- Blue Wire - Input 3 – up to 24VDC signal from Green Button

***** AMERICAN WELDOQUIP only provides the torch, torch buttons and wire connections. The Robot Control Box and any programming is supplied by the customer. *****

DAILY MAINTENANCE

A few minutes per day performing a quick check of your MIG torch will help to decrease weld problems, minimize downtime, and help increase consumable life.

At Beginning of Shift

- Inspect the cable for cuts, nicks, or tears. If you can see bare copper, return for maintenance.
- Inspect the front-end consumables. Clean weld spatter and inspect the nozzle insulator. If the nozzle insulation is damaged it should be replaced.
- Check that the gas diffuser is tight, and the insulating washer is on the gooseneck.

- The insulating washer is pertinent to have in place as this helps prevent the gooseneck from getting electrically charged.
- Check the gas holes in the diffuser and clean if necessary.
- Check and tighten the contact tip.
- Check all electrical connections including the power cable from the power supply, torch/feeder connections, and control cables for loose connections. Tighten if necessary. Loose connections can cause overheating of cables and/or loss of electrical power.

MAINTENANCE

LINER REPLACEMENT

Warnings – To avoid accidental injury ensure power supply and wire feed unit is turned off.

LINER REMOVAL

- 1) Trim the end of the weld wire at contact tip.
- 2) Retract or completely remove weld wire so torch can be removed from the wire feeder.
- 3) Remove the nozzle, contact tip and diffuser.
- 4) Remove the torch cable from the wire feed unit.
- 5) Loosen the set screw at the torch feeder connection using a 5/64" Allen wrench or remove the feeder pin liner cap.
- 6) Making sure the torch cable is straight, grasp the liner at the rear of the torch with a pair of pliers and pull to remove.

LINER INSTALLATION

- 1) Carefully feed the new liner into the torch using short strokes to avoid kinking.
- 2) Continue feeding the liner until the collet seats against the feeder pin.
- 3) Tighten the set screw using a 5/64" Allen wrench or resecure the feeder pin liner cap.
- 4) Reinstall the torch to the wire feed unit.
- 5) Trim the liner to ¾" stick out from the end of the gooseneck.
- 6) Replace the diffuser, contact tip and nozzle.
- 7) Feed welding wire into the torch and tighten drive rolls.



WARNING: WHEN FEEDING WELD WIRE THROUGH THE TORCH

KEEP THE FRONT END OF THE TORCH POINTED AWAY FROM ANY PERSON OR OBJECT. DO NOT POINT AT THE FACE, HANDS ETC. FAILURE TO DO SO WILL RESULT IN BODILY INJURY AND POSSIBLY DEATH.

GOOSENECK REPLACEMENT

GOOSENECK REMOVAL

- 1) Remove both plates from back side of the Bracket with the 0.050" Allen Wrench (supplied with the bracket). **Do not lose or discard the screws as they are VERY small.**
- 2) Disconnect the Control Box wires from the LED wires inside the Bracket.
- 3) Loosen the screw in the top of the Bracket with a 5mm Allen Wrench (not supplied).
- 4) Remove the Gooseneck/Torch from the Bracket.
- 5) Loosen the Cobot Collar from the Gooseneck with a 2mm Allen Wrench (not supplied) and pull off the front of the gooseneck being careful pulling over the wiring. Set aside for reuse.
- 6) Unscrew the Control Box from the Adaptor Support with a 5mm Allen Wrench (not supplied). Set the screws aside for reuse.

- 7) Remove the Adaptor Support Screw with a flat screwdriver (not supplied). **CAREFULLY** slide the adaptor support back on the cable. **Be careful not to damage the wiring.**
- 8) Secure the Gooseneck and using a 5/8" wrench, loosen the Gooseneck/Power Cable connection and remove from the Power Cable/Adaptor Support.

GOOSENECK INSTALLATION

- 1) Thread the new Gooseneck onto the Power Cable/Adaptor Support and fully tighten.
- 2) Slide the Adaptor Support up and onto the Gooseneck aligning the holes carefully pulling the wires of the Control Box out to keep from getting pinched.
- 3) Reattach the Adaptor Support Screw to connect the Adaptor Support to the Gooseneck.
- 4) Tuck all wiring back into the Adaptor Support and the Control Box while aligning the screw holes up and reattaching. Be careful not to pinch wiring under the Control Box. The bullet end red/white wires will go under the end of the control box in the channel. This will trap the wiring in place and direct it towards the gooseneck. Tighten down with the screws.
- 5) Slide the Cobot Collar back over the front end of the gooseneck. The bullet end red/white wires will go through the center and into the channel of the Cobot Collar. Slide the collar up against the Control Box aligning the changes of the Cobot Collar and the Control Box. Tighten the set screw of the Cobot Collar.
- 6) Slide the Gooseneck/Torch back into the Bracket aligning the flats on both items. Tighten the set screw in the top of the Bracket.
- 7) Reattach the Control Box wiring to the LED wiring in the Bracket.
- 8) Replace the plates on the Bracket.
- 9) Install all new Consumables onto the Gooseneck.

POWER CABLE REPLACEMENT

POWER CABLE REMOVAL

- 1) Remove the Gooseneck from the torch, see Gooseneck Removal.
- 2) Disconnect the Control Box from the Power Cable.
- 3) Slide the front Adaptor Support off the front of the Power Cable and set aside for reuse.
- 4) Remove the Liner from the torch, see Liner Removal.
- 5) Remove the Plug around the wiring on the rear Adaptor Support and set aside for reuse.
- 6) Unthread the rear Adaptor Spring on the rear Adaptor Support (will be in 2 pieces) and set aside for reuse.
- 7) Remove the Adaptor Support Screw from the rear Adaptor Support and slide the rear Adaptor Support up the power cable.
- 8) Using a 5/8" wrench, loosen the Power Cable/Feeder Pin Block connection. Remove the Feeder Pin Adaptor Block with Feeder Pin and set aside for reuse.
- 9) Slide the rear Adaptor Support and Spring off the new Power Cable.

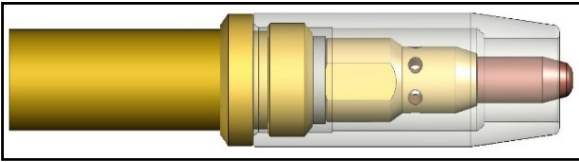
POWER CABLE INSTALLATION

- 1) Slide the Spring and rear Adaptor Support onto the new Power Cable. This end will have (4) wires w/ (4) bullets on the end.
- 2) Attach the Feeder Pin Adaptor Block with Feeder Pin onto the new Power Cable and fully tighten.
- 3) Attach the rear Adaptor Support onto the Feeder Pin Adaptor Block with Feeder Pin with the Adaptor Screw aligning the holes in each.
- 4) Slide the (4) wires w/ (4) bullets through the rectangular window of the rear Adaptor Block.
- 5) Reattach the plug into the rectangular window around the wiring.

- 6) Attach both pieces of the rear Adaptor Support together and screw the Spring back on to the rear Adaptor Support.
- 7) Reinstall the Gooseneck, see Gooseneck Installation.
- 8) Install a new Liner, see Liner Installation.

NOZZLE/TIP RELATIONSHIPS

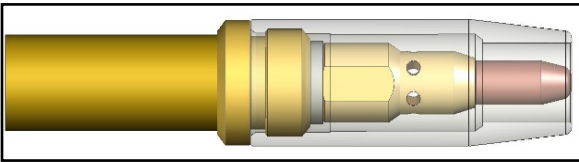
Shown below are typical relationship between the contact tip and nozzle in GMAW Semi-Automatic applications. Nozzles to tip relationships are usually dictated by the process and application but not necessarily the standard. Keep in mind that decreased tip life, increased spatter cleaning cycles may be required if the tip relationship is changed to achieve other objectives.



Extended Tip – Short Circuit (Short Arc, Dip Transfer) Welding Applications

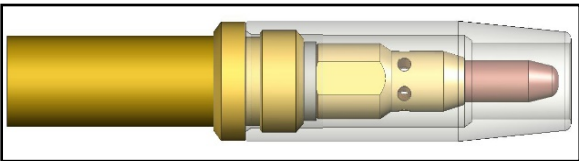
The tip stickout is usually 1/8" or 1/4" from the nozzle. Keep in mind that the further the stickout the more susceptible to gas porosity issues.

Typically, is used in short circuit, lower amperage applications, and/or where you may need to reach into a corner.



Flush Tip – Higher current Short Circuit (Short Arc, Dip Transfer) Welding Applications

The tip is flush with the end of the nozzle. Typically used in higher current and voltage short circuit applications.



Recessed Tip – Spray Arc, Pulsed, Flux Core Welding Applications

The tip is usually recessed in the nozzle 1/8" or 1/4". Usually, the higher the heat and/or current the further the recess. However, this is also

dependent on the wire used and the arc length requirement.

NOZZLE SELECTION CHART

300 SERIES NOZZLES

(300R SERIES)

<u>Part Number</u>	<u>Description</u>	<u>Bore Size</u>	<u>Material</u>	<u>O.A.L</u>	<u>Insulator</u>
75133801	CONICAL	3/8" (9.5mm)	NI PLATED BRASS	2.820" (71.6mm)	NONE REQ
75133802	BOTTLE NOSE	3/8" (9.5mm)	NI PLATED BRASS	2.820" (71.6mm)	NONE REQ
75133803	BOTTLE NOSE	3/8" (9.5mm)	NI PLATED BRASS	3.620" (91.9mm)	NONE REQ
75135004	SHORT CONICAL	1/2" (12.7mm)	NI PLATED BRASS	1.937" (49.2mm)	NONE REQ
75135002-I	TAPERED	1/2" (12.7mm)	NI PLATED BRASS	2.820" (71.6mm)	INSTALLED
75135003-I	TAPERED	1/2" (12.7mm)	NI PLATED BRASS	2.672" (68mm)	INSTALLED
75135601-I	CONICAL	9/16" (14.2mm)	NI PLATED BRASS	2.820" (71.6mm)	INSTALLED
75135602-I	CONICAL	9/16" (14.2mm)	NI PLATED BRASS	2.545" (64.6mm)	INSTALLED

400 SERIES NOZZLES

(400R – 500R SERIES)

<u>Part Number</u>	<u>Description</u>	<u>Bore Size</u>	<u>Material</u>	<u>O.A.L</u>	<u>Insulator</u>
75144301-I	TAPERED	7/16" (11.1mm)	NI PLATED BRASS	2.820" (71.6mm)	INSTALLED
75144302-I	TAPERED	7/16" (11.1mm)	NI PLATED BRASS	3.125" (79.4mm)	INSTALLED
75145001	BOTTLE NOSE	1/2" (12.7mm)	NI PLATED BRASS	3.031" (77.0mm)	75001733
75145002C	TAPERED	1/2" (12.7mm)	NI BARE COPPER	3.125" (79.4mm)	75001738
75145002CU	TAPERED	1/2" (12.7mm)	BARE COPPER	3.125" (79.4mm)	75001738
75145003	LONG TAPERED	1/2" (12.7mm)	NI PLATED BRASS	3.915" (99.4mm)	NONE REQ
75145004	BOTTLE NOSE	1/2" (12.7mm)	NI PLATED BRASS	3.125" (79.4mm)	75001733
75145005	BOTTLE NOSE	1/2" (12.7mm)	NI PLATED BRASS	2.820" (71.6mm)	75001733
75145601-I	TAPERED	9/16" (14.2mm)	NI PLATED BRASS	3.125" (79.4mm)	INSTALLED
75145602-I	SMALL CONICAL	9/16" (14.2mm)	NI PLATED BRASS	2.820" (71.6mm)	INSTALLED
75145603-I	SHORT TAPERED	9/16" (14.2mm)	NI PLATED BRASS	2.375" (60.3mm)	INSTALLED
75146202	CONICAL	5/8" (15.9mm)	NI PLATED BRASS	3.125" (79.4mm)	75001738
75146202C	CONICAL	5/8" (15.9mm)	NI PLATED COPPER	3.125" (79.4mm)	75001738
75146202CU	CONICAL	5/8" (15.9mm)	BARE COPPER	3.125" (79.4mm)	75001738
75146202-I	BOTTLE NOSE	5/8" (15.9mm)	NI PLATED BRASS	3.125" (79.4mm)	INSTALLED
75146204CU	CONICAL	5/8" (15.9mm)	BARE COPPER	2.820" (71.6mm)	75001738
75146204-I	CONICAL	5/8" (15.9mm)	NI PLATED BRASS	2.820" (71.6mm)	INSTALLED
75146205CU	TAPERED	5/8" (15.9mm)	BARE COPPER	3.031" (77.0mm)	75001738
75146205-I	CONICAL	5/8" (15.9mm)	NI PLATED BRASS	3.031" (77.0mm)	INSTALLED
75146206-I	SHORT CONICAL	5/8" (15.9mm)	NI PLATED BRASS	1.875" (47.6mm)	INSTALLED
75146207-I	SHORT CONICAL	5/8" (15.9mm)	NI PLATED BRASS	2.187" (55.6mm)	INSTALLED
75146501S	SPOT	5/8" (15.9mm)	NI PLATED BRASS	3.437" (87.3mm)	75001738
75147501	CYLINDRICAL	3/4" (19.0mm)	NI PLATED BRASS	3.125" (79.4mm)	75001738
75147501CU	CYLINDRICAL	3/4" (19.0mm)	BARE COPPER	3.125" (79.4mm)	75001738

CONTACT TIP SELECTION CHART

M6 CONTACT TIPS

(300R SERIES)

<u>Part Number</u>	<u>Wire Size</u>	<u>Nominal I.D</u>
Copper (CU)		
75023511	0.023" (.6mm)	0.034"
75030511	0.030" (.8mm)	0.038"
75035511	0.035" (.9mm)	0.044"
75040511	0.040" (1.0mm)	0.048"
75045511	0.045" (1.2mm)	0.054"

M8 CONTACT TIPS

(400R – 500R SERIES)

<u>Part Number</u>	<u>Wire Size</u>	<u>Nominal I.D</u>
Copper (CU)		
75030014	0.030" (.8mm)	0.038"
75035014	0.035" (.9mm)	0.044"
75040014	0.040" (1.0mm)	0.048"
75045014	0.045" (1.2mm)	0.054"
75047014 (FOR ALUMINUM)	0.045" (1.2mm)	0.057"
75052014	0.052" (1.3mm)	0.061"
75062014	0.062" (1.6mm)	0.073"
75062015 (FOR ALUMINUM)	0.062" (1.6mm)	0.076"
75078014	0.078" (2.0mm)	0.087"
75093014	0.093" (2.4mm)	0.106"
Heat Resistant Technology (HRT) – Standard Length		
20030400	0.030" (.8mm)	0.039"
20035400	0.035" (.9mm)	0.041"
20040400	0.040" (1.0mm)	0.050"
20045400	0.045" (1.2mm)	0.054"
20052400TT	0.052" (1.3mm)	0.057"
20062400	0.062" (1.6mm)	0.069"
Heat Resistant Technology (HRT) – Short Length		
20030500	0.030" (.8mm)	0.039"
20035500	0.035" (.9mm)	0.041"
20040500	0.040" (1.0mm)	0.050"
20045500	0.045" (1.2mm)	0.054"
20052500	0.052" (1.3mm)	0.061"
20062500	0.062" (1.6mm)	0.069"

*TT is for tight tolerance.

TEACH TIPS

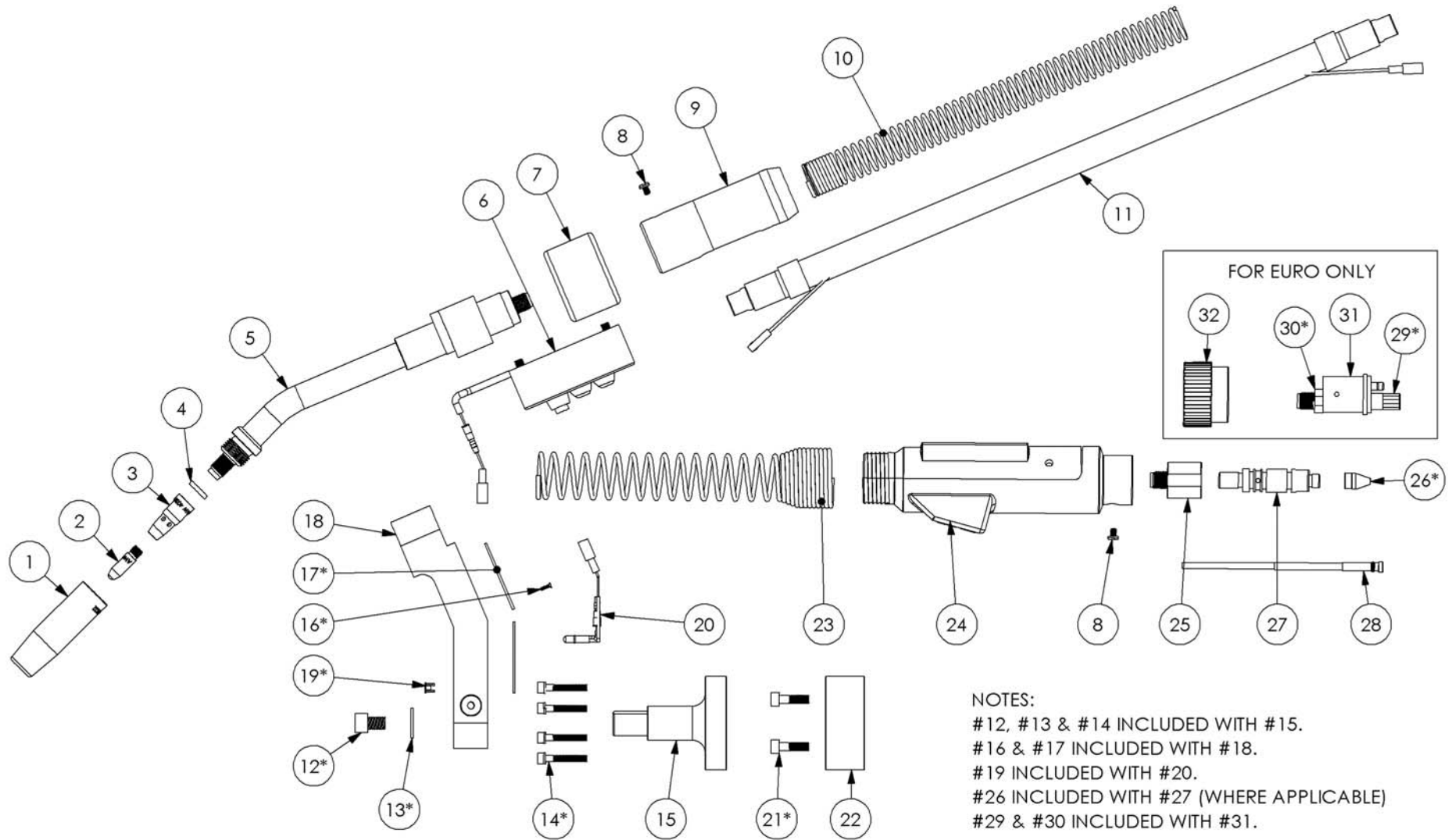
<u>Series</u>	<u>Part Number</u>	<u>Description Tip Series/Wire Stickout</u>	<u>Material</u>
300	750TT511-.472	511 tip with 0.472 (11.9mm) Stickout	Steel
300	750TT511-.590	511 tip with 0.590 (15mm) Stickout	Steel
300	750TT511-.625	511 tip with 0.625 (15.9mm) Stickout	Steel
400/500	200TT230-.625	230/500 tip with 0.625 (15.9mm) Stickout	Steel
400/500	750TT014-.472	014/400 tip with 0.472 (11.9mm) Stickout	Steel
400/500	750TT014-.500	014/400 tip with 0.500" (12.7mm) Stickout	Steel
400/500	750TT014-.590	014/400 tip with 0.590" (15mm) Stickout	Steel
400/500	750TT014-.625	014/400 tip with 0.625" (15.9mm) Stickout	Steel
400/500	750TT014-.750	014/400 tip with 0.750" (19mm) Stickout	Steel
400/500	750TTP014	014/400 tip w/ 0.590", 0.625", 0.750" Stickout	Plastic

LINER SELECTION CHART

(ALL SERIES AND FEEDER PINS)

<u>Part Number</u>	<u>Description</u>	<u>Length</u>	<u>Material</u>
For Steel Wire Applications			
75005215T	0.023-0.030 WIRE LINER	5FT	STEEL
75010215T	0.023-0.030 WIRE LINER	10FT	STEEL
75015215T	0.023-0.030 WIRE LINER	15FT	STEEL
75001222T	0.035-0.045 WIRE LINER	1FT	STEEL
75005222T	0.035-0.045 WIRE LINER	5FT	STEEL
75010222T	0.035-0.045 WIRE LINER	10FT	STEEL
75015222T	0.035-0.045 WIRE LINER	15FT	STEEL
75025222T	0.035-0.045 WIRE LINER	25FT	STEEL
75001228T	0.045-1/16 WIRE LINER	1FT	STEEL
75005228T	0.045-1/16 WIRE LINER	5FT	STEEL
75010228T	0.045-1/16 WIRE LINER	10FT	STEEL
75015228T	0.045-1/16 WIRE LINER	15FT	STEEL
75025228T	0.045-1/16 WIRE LINER	25FT	STEEL
75001229T	5/64-3/32 WIRE LINER	1FT	STEEL
75005229T	5/64-3/32 WIRE LINER	5FT	STEEL
75010229T	5/64-3/32 WIRE LINER	10FT	STEEL
75015229T	5/64-3/32 WIRE LINER	15FT	STEEL
75025229T	5/64-3/32 WIRE LINER	25FT	STEEL
For Aluminum Wire Applications			
75005320T-J	0.035-0.045 WIRE LINER	5FT	TEFLON
75010320T-J	0.035-0.045 WIRE LINER	10FT	TEFLON
75005324T-J	0.045-1/16 WIRE LINER	5FT	NYLON
75010324T-J	0.045-1/16 WIRE LINER	10FT	NYLON
75015324T-J	0.045-1/16 WIRE LINER	15FT	NYLON

PARTS BREAKDOWN



NO	DESCRIPTION	AIR COOLED		WATER COOLED
		300R	500R	550R
1	NOZZLE	SEE NOZZLE CHART		
2	CONTACT TIP	SEE TIP CHART		
3	GAS DIFFUSER, STD	75002007	75004033	
	GAS DIFFUSER, SHORT	75002007S	75004033S	
	GAS DIFFUSER, LONG	75002007L	75004033L	
	GAS DIFFUSER, X LNG	7502007XL	75004033XL	
4	INSULATING WASHER	75001003	75001004	
5	GOOSENECK 22°	300R-22-H	500R-22-H	550R-22-H
	GOOSENECK 45°	300R-45-H	500R-45-H	550R-45-H
6	COBOT CTRL BOX KIT	CB-1000		
7	COBOT COLLAR	CB-0009		CB-0009W
	COBOT COLLAR FOR 45° GOOSENECK	CB-0009-45		CB-0009W-45
8	ADAPTOR SCREW	75077011		
9	FRONT ADAPTOR	75077017-CB		75077017W-CB
	FRONT ADAPTOR FOR 45° GOOSENECK	75077017-CB45		75077017W-CB45
10	CABLE SUPPORT SPRING	75002009		
11	10FT POWER CABLE	75010300R-CB	75010500R-CB	
	12FT POWER CABLE	75012300R-CB	75012500R-CB	
NS	WATER HOSE CLAMP	NOT REQUIRED		38577105
NS	WATER HOSE	NOT REQUIRED		75099025
NS	MALE QUICK DISCONNECT FITTING	NOT REQUIRED		38660000
12*	ARCSAFE™ BRACKET SCREW	37677119		
13*	ARCSAFE™ BRACKET WASHER	901644		
14*	ARCSAFE™ BRACKET MOUNT SCREWS	37677120		
NS*	ARCSAFE™ BRACKET MOUNT WASHERS	RQ-0508		
15	ARCSAFE™ SOLID MOUNT	AS-B-A-A		
16*	ARCSAFE™ COBOT PLATE SCREW	37677134		
17*	ARCSAFE™ COBOT BRACKET PLATE	CB-0010		
NS*	COBOT BRACKET PLATE ALLEN WRENCH	0.05HEXKEY		
18	ARCSAFE™ 23° COBOT BRACKET	AS-B-23-CB		
19*	ARCSAFE™ LED HOUSING CLIP	CB-0007		
20	ARCSAFE™ COBOT LED WIRE HARNESS	CB-1001		
21*	ARCSAFE™ INSULATING DISC SCREWS	SEE INSULATING DISC CHART		
22	ARCSAFE™ INSULATING DISC	SEE INSULATING DISC CHART		
23	REAR ADAPTOR SPRING	SEE ADAPTOR	75002018	
24	REAR ADAPTOR	75077004-CB	75077006	
25	REAR ADAPTOR BLOCK	37077003		
26*	LINER CAP - FRONIUS (FSC)	*SEE APPENDIX A-1*		
	LINER CAP - MILLER	37577705-N		
	LINER CAP - OTC	38277001-N078		
NS*	O-RING – EURO	75000002		
NS*	O- RING- FRONIUS (FSC)	75000057		
NS*	O-RING – MILLER	37577102		
NS*	O-RING – TWECO #4	37677102		

NO	DESCRIPTION	AIR COOLED		WATER COOLED
		300R	500R	550R
NS*	O-RING – TWECO #5	75000021		
27	FRONIUS FSC FEEDER PIN	37677711*SEE APPENDIX A-1*		
	LINCOLN LN7 FEEDER PIN KIT	37677832		
	LINCOLN UNIVERSAL FEEDER PIN KIT	37677837		
	MILLER FEEDER PIN	37577705		
	OTC / DAIHEN FEEDER PIN	37577700		
	OTC DP SERIES FEEDER PIN	37577701		
	PANASONIC FEEDER PIN	37577999		
	TWECO #4 FEEDER PIN	37577699		
	TWECO #5 FEEDER PIN	37577930		
28	LINER	SEE LINER CHART		
29*	LINER CAP - EURO	37577005		
30*	EURO TO POWER CABLE ADAPTOR	75001151		
31	EURO ADAPTOR BLOCK	75001147		
32	EURO FEEDER NUT	75077014		

** All Feeder Pin Kits include their gas hose kits, where applicable. To get just the feeder pin from that kit, the P/N starts with 375xxxxx. (Except Euro and Fronius FSC)

** For Feeder Pins, item #25 not needed for Euro and Fronius FSC Feeder Pins.

** For additional information, please contact **AMERICAN WELDQUIP**.

INSULATING DISK CHART

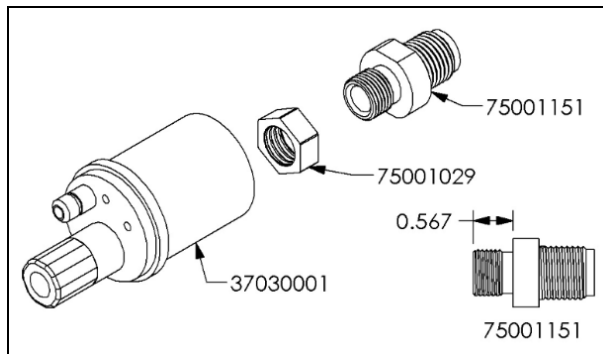
<u>Robot Make</u>	<u>Model</u>	<u>Insulating Disc</u>	<u>Robot Mounting Screws</u>
ABB	GoFa CRB 15000	AS-250-A	(4) 37677127
Aubo	i10, CB4	AS-250-A	(4) 37677127
Fanuc	CRX-10iA, 10iA/L	AS-250-A	(4) 37677127
Universal	UR5, UR10, UR10e	AS-250-A	(4) 37677127

** For additional robot makes and models, please contact **AMERICAN WELDOQUIP**.

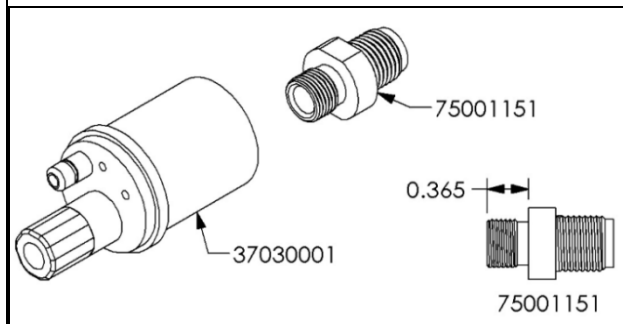
APPENDIX

A-1 EURO FEEDER PIN

Before January 2024
No longer Available



January 2024 and after
P/N 75001147

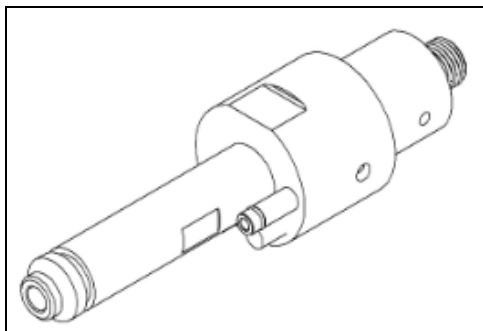


If 75001151 length is 0.567" long, P/N 75001029 is still needed and is still available. 37030001 will come with pins and wires used in hand-held applications. Not used in this application.

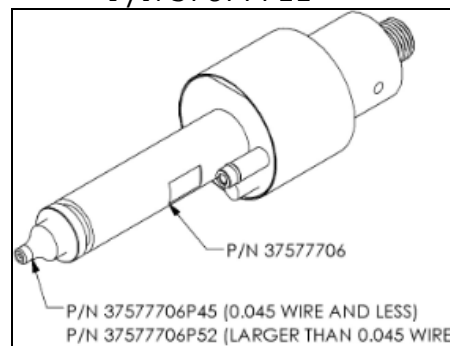
When ordering 75001147, 37030001 will not have pins and wires.

A-2 FRONIUS FSC FEEDER PIN

Before February 2024
No longer Available



February 2025 and after
P/N 37677711



Switching from old style to new style. A new Pin (P/N 37577706) and Plug (37577706P45 [0.045 wire and smaller] or 37577706P52 [larger than 0.045 wire]) is needed. No need to purchase the entire assembly.

NOTES



AMERICAN WELDQUIP

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Sharon Center, Ohio 44274

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Fax 800-949-9353

www.weldquip.com

www.youtube.com/weldquip1



OM023 – 05/25